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# Magnetic polarity stratigraphy and possible differential tectonic rotation of the Miocene–Pliocene mammal-bearing San Timoteo Badlands, southern California

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**Abstract**

The San Timoteo Badlands, consisting of the Mt. Eden beds conformably overlain by the San Timoteo beds, is a 3000 m thick sequence of nonmarine sediments deposited in a tectonically active region in southern California. Paleomagnetic samples were collected from 132 sites within these badlands. A composite magnetostratigraphic section for this sequence correlates to the MPTS from the late part of chron C3An.1n to the early part of chron C2r.2r, between about 6 and 2.3 Ma. The classic late Hemphillian (late Miocene) Mt. Eden fossil mammal quarry occurs near the base of the composite section within chron C3r, between 5.8 and 5.3 Ma. Although the overall unit-mean declinations indicate negligible or slight counterclockwise (CCW) vertical-axis rotation ($-8.8° \pm 4.9°$), analysis of declination subsets from different parts of the section suggest possible differential small-block rotation within the San Timoteo Badlands: Prior to 3.5 Ma (represented by the Mt. Eden beds) there is negligible rotation; the overlying portion of the San Timoteo Badlands (represented by the lower San Timoteo beds) dated between 3.5 and 2.3 Ma (chrons C2An.3n to C2r.2r), indicate significant apparent CCW rotation ($R \pm \Delta R$) of $-20.7° \pm 5.2°$. These data suggest decoupling of the upper part of the San Timoteo Badlands and independent CCW vertical-axis rotation relative to the lower part of this sequence.

*Keywords:* San Timoteo Badlands; magnetostratigraphy; neotectonics; Pliocene; Miocene

## 1. Introduction

The San Timoteo Badlands consist of a 3000 m thick sequence of nonmarine Miocene–Pleistocene deposits located in Riverside County, southern California (Fig. 1A). These badlands consist primarily of two lithostratigraphic units: (1) the underlying Mt. Eden beds, which contain late Miocene land mammals from the classic Mt. Eden fossil quarry [1]; and (2) the conformably overlying San Timoteo beds, which contain localized occurrences of Pliocene to Late Pleistocene land mammals. These badlands are important for North American land mammal biostratigraphy because they represent one of the only places where a terrestrial sedimentary sequence

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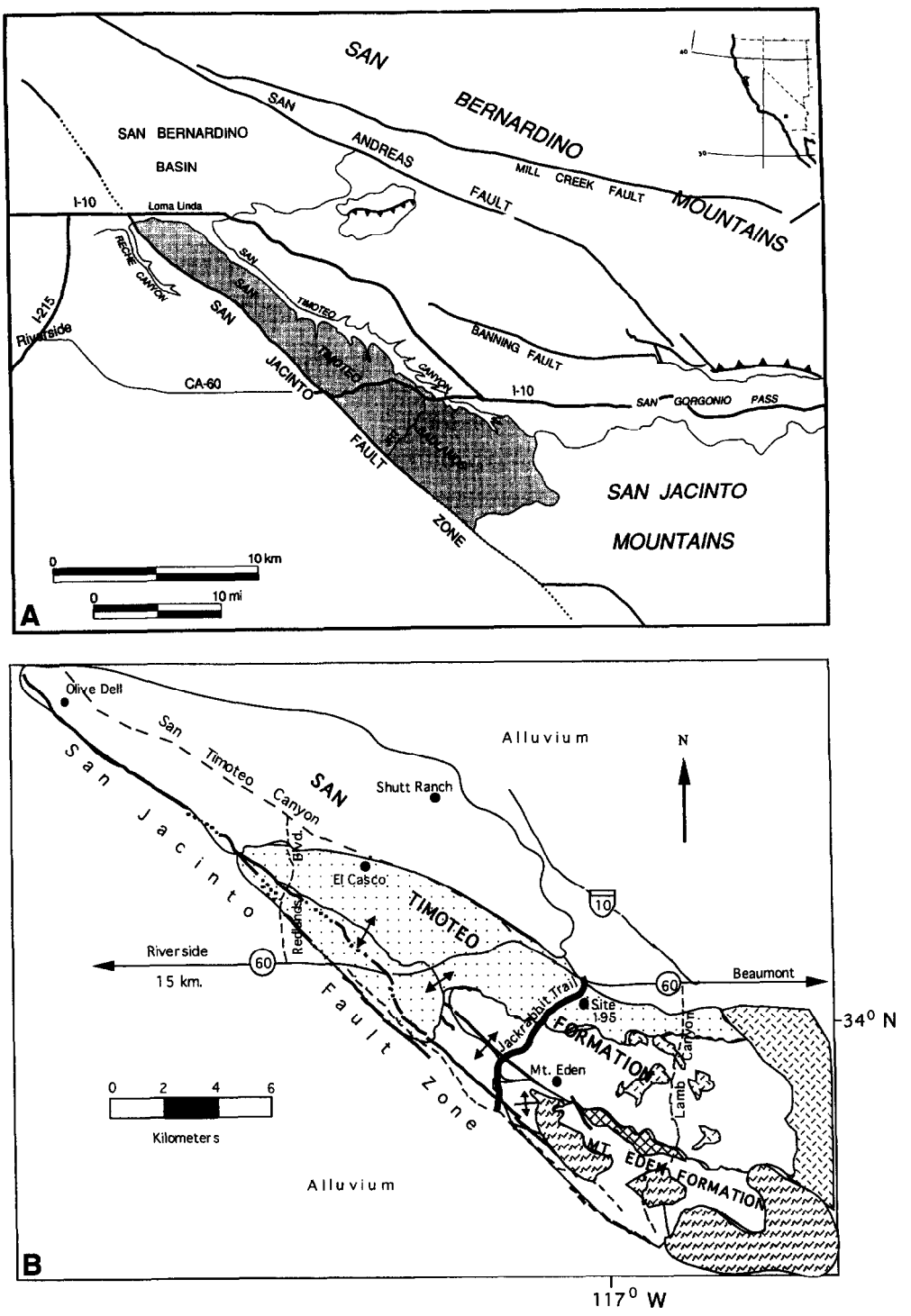



Fig. 1. (A) Map of the San Timoteo Badlands region including the major fault systems. (B) Location of sections sampled within the badlands. ● = principal mammal-bearing fossil localities (discussed in this paper and known from other studies). Our principal measured section and paleomagnetic transect was taken along Jackrabbit Trail.

spans, in demonstrable superposition, an age from about 6 to 0.5 Ma.

With regard to their tectonic significance, the San Timoteo Badlands are bounded on the southwest by an active branch of the San Andreas fault system, the San Jacinto fault zone (Fig. 1). The Banning fault zone and the south and north branches of the San Andreas fault lie approximately 10 km to the north (Fig. 1A). As such, the section is ideally situated to clarify the slip history of this segment of the San Andreas fault system and to provide information concerning the uplift of the southern front of the San Bernardino Mountains.

Previous studies have provided a geologic and stratigraphic framework for the region [2–4]; however, there have been no prior detailed analyses of the age and correlation of the San Timoteo Badlands. Accordingly, the purpose of this paper is to present new magnetostratigraphic results of the classic Hemphillian (late Miocene) Mt. Eden fossil mammal locality and the long and relatively continuous stratigraphic section containing the Mt. Eden and San Timoteo beds exposed along the Jackrabbit Trail, a road that runs perpendicular to strike across the badlands anticline (Fig. 1B). This is the first part of a combined effort to elucidate the Late Miocene to middle Pliocene paleontology and geological history of this tectonically active area. Other work in progress by our group will subsequently address questions in the upper part of this sequence (middle Pliocene to Middle Pleistocene) exposed northwest of Jackrabbit Trail and north of State Route 60 (Fig. 1) as new fossil sites are described and integrated with other measured sections and paleomagnetic sampling transects.

## 2. Geographic location, geologic setting, and previous studies

The San Timoteo Badlands are located in Riverside County, ca. 25–30 km east of Riverside at lat. 33°55′ N and long. 117°3′ W (Fig. 1). The sections sampled and field study area are contained within the USGS 7.5 minute series El Casco Quadrangle topographic map.

The badlands sedimentary sequence consists of the basal Mt. Eden beds conformably overlain by the San Timoteo beds. The total section sampled during the current study consists of approximately 700 m, including 50–75 m of the Mt. Eden formation (informal unit) and 475–500 m of the San Timoteo 'formation' (informal stratigraphic unit; the Mt. Eden and San Timoteo deposits have not yet been formally described) exposed along the Jackrabbit Trail, south of State Route 60. Another important section studied here is 150 m thick, located about 0.7–0.8 km east of Jackrabbit Trail, and includes the late Hemphillian Mt. Eden fossil mammal quarry.

### 2.1. Stratigraphy

The Mt. Eden deposits exposed along the Mt. Eden measured sections and the lower 60 m of Jackrabbit Trail consist of lacustrine siltstones, claystones, carbonate mudstones, and shales. The upper 50 m of the Mt. Eden beds exposed along Jackrabbit Trail consists of fluvial siltstones, sandy siltstones, and ripple-laminated, fine-grained sandstones. As will also be described below, the Mt. Eden sediments contain a diverse late Hemphillian (latest Miocene) land mammal faunal assemblage [1]. The overlying San Timoteo beds consist of fluvial siltstones, silty sandstones, sandstones with interbedded pebbly sandstones, conglomerates, and contain localized Pliocene to mid-Pleistocene land mammals.

### 2.2. Tectonic / structural setting and previous geological studies

The San Timoteo Badlands are bounded on the southwest by the San Jacinto fault zone, the most active branch of the San Andreas fault system in this region. The Banning fault zone and the north and south branches of the San Andreas fault zone lie, respectively, 5 km and 10 to 15 km north of the badlands (Fig. 1A). The sediments of the Mt. Eden and San Timoteo formations are exposed in an anticline that plunges gently to the northwest. Jackrabbit Trail, which trends to the northeast, contains continuous exposures of approximately one-third of the entire San Timoteo Badlands sequence.

Matti and Morton [2] determined that the basal Mt. Eden sediments contain clasts indicating derivation from Peninsula Range basement highs, such as Mt. Eden (locally) and the San Jacinto Mountains (to

the east). This lowermost arkosic unit interfingers with the overlying lacustrine package. Approximately 4 myr ago, "a large alluvial fan and braided stream complex consisting of sediment derived from the Transverse Ranges [began] to prograde southward onto the Peninsula Ranges block" [2] to form the fluvial San Timoteo deposits that conformably overlie the Mt. Eden beds. Southwestward, paleocurrent indicators in thick conglomeratic lenses throughout the San Timoteo sediments support this model [3,4]. After about 1.5 Ma, rock types indicative of a San Bernardino Mountains source appear in the San Timoteo deposits [5,6]. The chronology developed during this project will refine our understanding of the changing source regimes as well as uplift and emplacement of the San Bernardino Mountains.

### 2.3. Previous paleontological studies and biostratigraphic correlations

Fossil land mammals from the San Timoteo Badlands were first described in 1921 by Childs Frick [1]. He documented a large and diverse assemblage of Hemphillian (late Miocene) age. Although the now-classic Mt. Eden Local Fauna (L.F.) has since been mentioned as a biogeographically important assemblage of late Miocene age [7,8], until the 1980s paleontological studies, or additions to this fauna, as well as from other parts of the San Timoteo Badlands, were few. After this hiatus, a biostratigraphic study of the Mt. Eden beds and more precise correlation to the late Hemphillian was based on the presence of diagnostic taxa, including the rodent *Repomys* and horse '*Pliohippus*' (= *Dinohippus*) [9]. May and Repenning [9] also presented preliminary paleomagnetic data in which they obtained reversed polarities from two sites. Based on the biostratigraphic presence of diagnostic late Hemphillian mammals and reversed polarity, they correlated one of the original Frick quarries at Mt. Eden to the earliest reversed zone (chron 3r, *sensu* Berggren et al. [10]) within the Gilbert Chron, with an age between 5.8 and 5.3 Ma.

The long and continuous stratigraphic section along Jackrabbit Trail (JRT), which contains some of the best exposures of the San Timoteo Badlands, is generally not fossiliferous. Based exclusively on lithology, the lower part of the section at JRT correlates to the short section containing the Mt. Eden fossil quarry. While the lower part of the JRT section lacks fossils, a few specimens, including the relatively primitive equine horse '*Equus*' (*Plesippus*) (*sensu* [11]), indicating a Blancan age, have been collected from the top of this section. Given the known age range for primitive *Equus*, this occurrence is early to middle Pliocene, certainly younger

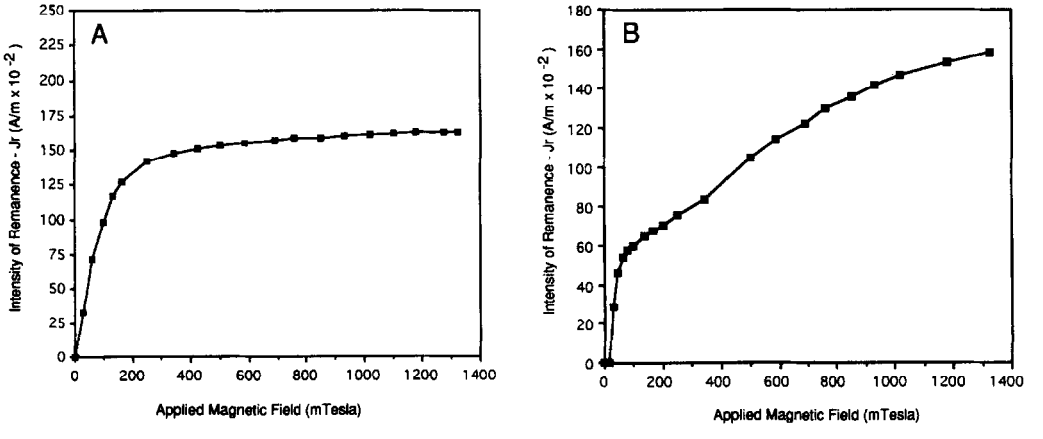



Fig. 2. Isothermal remanent magnetization (IRM) acquisition curves for selected paleomagnetic samples analyzed during this study. (A) Typical IRM acquisition curve for samples dominated by a low-coercivity magnetic mineral, probably magnetite. Saturation is achieved in applied fields of less than 200 mT. (B) Typical IRM acquisition curve for samples dominated by a high-coercivity magnetic mineral, probably hematite. Saturation is not achieved in applied fields of up to 1,400 mT (and ranging up to 3,500 mT in some cases; not shown here).

than the Hemphillian–Blancan boundary at ca. 4.5 Ma [12], and could be as young as ca. 2.5–2.0 Ma.

## 3. Paleomagnetic sampling, laboratory methods, statistical analysis and results

### 3.1. Paleomagnetic sampling

Paleomagnetic samples were collected from the short section that includes the Mt. Eden L.F. and the extensive section along Jackrabbit Trail. These sections provide a relatively composite section, representing the badlands sequence from the Mt. Eden through to the middle part of the San Timoteo beds. As is standard field procedure [13], three separately oriented paleomagnetic samples were collected at each of 43 sites in the Mt. Eden section and 80 sites along Jackrabbit Trail. Sites were spaced 2–5 m apart stratigraphically in the Mt. Eden lacustrine deposits. Site spacing was increased to 10–15 m for the fluvial upper Mt. Eden and San Timoteo succession. Some variation in site spacing was necessary to obtain suitably fine-grained lithologies and to avoid obviously faulted, disturbed, or highly altered sediments.

### 3.2. IRM treatment

In order to determine an effective regime of laboratory demagnetization, 15 samples from throughout the section and representative of the varied lithologies present, were subjected to isothermal remanence (IRM) studies to determine the dominant magnetic minerals. For 13 of the 15 samples, saturation was achieved in magnetizing fields of less than 1000 milliTesla (mT; e.g., Fig. 2A). This saturation magnetization behavior suggests magnetite is the dominant carrier of the natural remanent magnetization (NRM). In addition, paleomagnetic sample behavior during thermal demagnetization suggests that the slight increases seen in the IRM results for applied fields above 200 mT are most likely attributable to the presence of goethite, a high coercivity, low unblocking temperature magnetic mineral. Two of the 15 samples exhibited saturation magnetization behavior, suggesting hematite as the dominant magnetic mineral (Fig. 2B). Saturation did not occur at

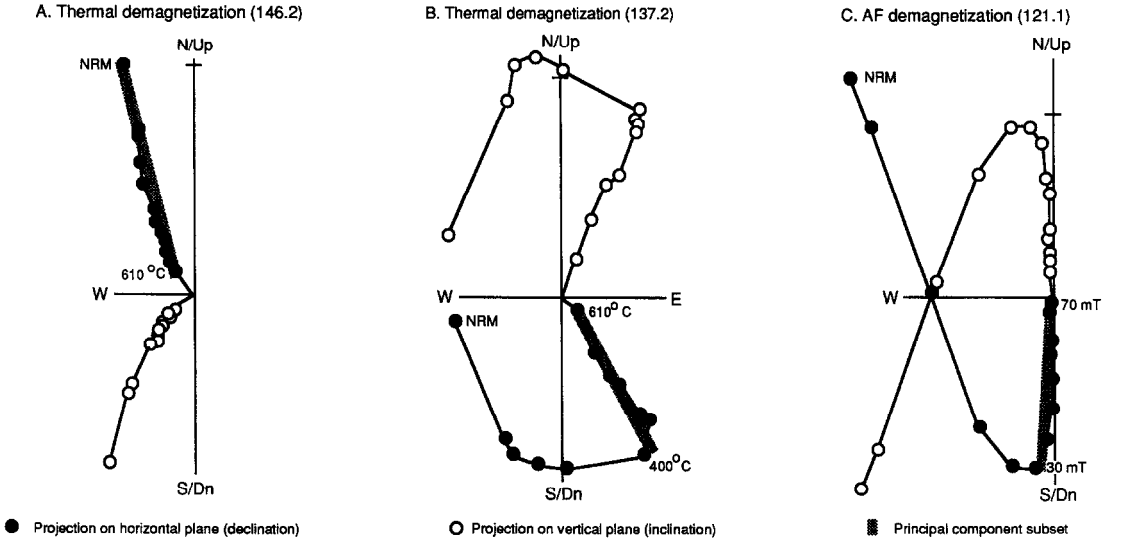



Fig. 3. Representative vector component demagnetization (Zijderveld) diagrams showing typical thermal demagnetization behavior for (A) normally magnetized and (B) reversely magnetized samples. For comparison, (C) shows typical alternating field demagnetization behavior of a reversely magnetized sample. Characteristic directions interpreted from principal component analysis are indicated by shaded traces of horizontal component.

applied fields up to 3500 mT. The inflection in applied field of less than 200 mT indicates the presence of a low-coercivity mineral, probably magnetite.

### 3.3. Magnetic laboratory analysis

After measurement of the NRM, each sample was incrementally treated, using either thermal and/or alternating field (AF) demagnetization, and measured on a 2G cryogenic magnetometer located in a magnetically shielded room in the University of Florida Paleomagnetics Laboratory. The samples were demagnetized to complete destruction (i.e., $< 1\%$ of the NRM intensity), 610°C, or 75 mT.

The samples exhibiting stronger initial magnetization were typically demagnetized in 100° steps up to 400°C, 50° or 25° steps up to 600°C, and 10° steps up to 610°C or 620°C. Samples displaying moderate initial remanent magnetization were demagnetized using a combination of thermal and AF demagnetization, ending with thermal demagnetization steps of 25°C until destruction of the remanent magnetism.

### 3.4. Statistical treatment and results

Data resulting from thermal and AF demagnetization were plotted on orthogonal vector plots [14]. Some samples exhibit single component behavior (Fig. 3A). However, the NRM of most samples is characterized by two components (Fig. 3B,C), one of which is a low-coercivity, viscous overprint in the direction of the present field. In either case, higher temperature demagnetization ($> 300$°C) usually yielded a stable, characteristic component, presumably carried by magnetite and, less often, hematite, which decayed toward the origin.

To determine characteristic paleomagnetic directions, principal component analysis was performed for each sample. Linear segments of the orthogonal plots were chosen by visual inspection and a least-squares best-fit line was calculated for each [15]. For two samples exhibiting poorly constrained principal components, the final direction was chosen using a single high-temperature step, usually between 500° and 610°C. The resulting paleomagnetic directions were then analyzed using Fisher statistics [16] to determine site mean declinations, mean inclinations, and associated statistical parameters.

All sites were then classified according to Opdyke et al. [17]. Class I sites ($N = 91$) are those with three samples with statistically significant ($R > 2.62$), concordant directions. Class III sites ($N = 23$) have three samples available, one of which exhibits a discordant direction. Of the 123 originally sampled sites, 4 were lost during cutting or transport and 5 sites are considered to be Class IV, either because the characteristic directions are highly scattered within the site or because only one sample from the site was available for analysis. As such, the laboratory analyses resulted in 93% of the sites ultimately available for magnetostratigraphic interpretations.

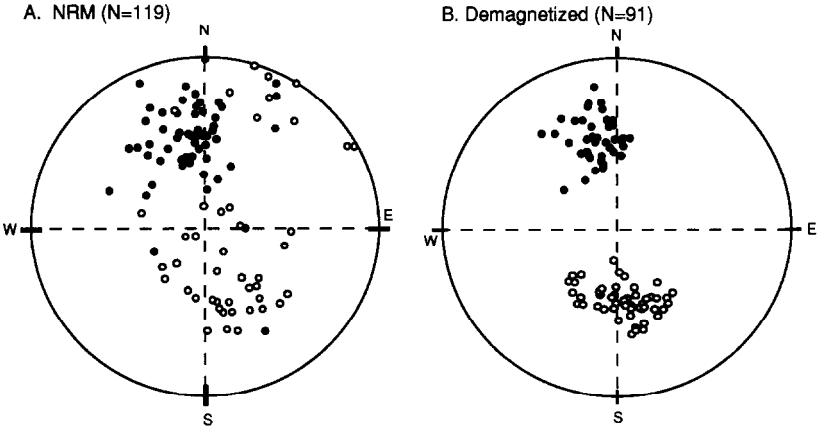



Fig. 4. Site mean directions plotted on equal-area stereographic projections. (A) Before demagnetization for all 119 sites available for analysis. (B) After either AF or thermal demagnetization for the 91 Class I sites.

Sites designated Class IV were interpreted as being either of normal or reversed polarity using the information available from the one sample, or by giving additional weight to inclination (in contrast to declination) data. These sites, designated as such, have been included in their respective positions in the paleomagnetic transects; however, they are considered to be statistically unreliable and therefore not used for polarity or tectonic interpretations.

Unlike the untreated (NRM) data, after demagnetization and statistical analysis, the 91 Class I site means grouped into either a northern- or southern-hemisphere population which are interpreted to represent, respectively, normal and reversed polarities (Fig. 4). The mean of all sites (reversed inverted to northern hemisphere) is dec. 350.9°, inc. 46.2° ($N = 91$, $\alpha_{95} = 3.1°$, $k = 27.4$). The mean of normal sites is dec. 346.8°, inc. 41.8° ($N = 40$, $\alpha_{95} = 4.1°$, $k = 35.6$), while the mean of all reversed sites is dec. 354.8°, inc. 50.4° ($N = 51$, $\alpha_{95} = 4.2°$, $k = 28.0$; Fig. 5). The difference between the mean directions of the normal and reversed polarity populations seems to result from two factors: (1) possible incomplete removal of a secondary overprint; (2) more importantly, a block of nine reversed polarity sites exhibiting large clockwise (CW) declination anomalies (rotation of 30–35°) accounts for a majority of the difference. Since no evidence of significant alteration (or of a normal overprint carried by hematite) is evident in these samples, this group of CW declination anomalies is probably attributable to local deformation (also see below). Excluding these nine anomalous sites, the overall mean of all sites (reversed polarity inverted) for the Mt. Eden–San Tim-

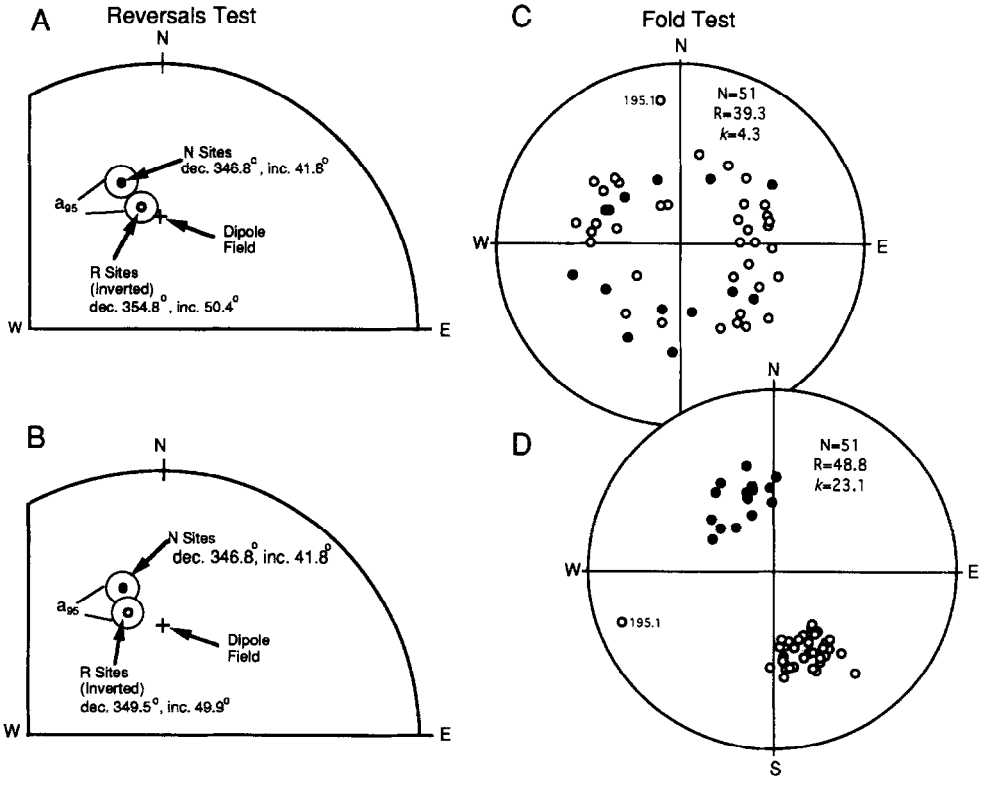



Fig. 5. Reversals (A) and (B) and fold (C) and (D) tests of paleomagnetic stability. (A) includes all 91 Class I sites. In (B) the group of nine reversed sites exhibiting anomalous CW rotation of about 30° has been removed (see discussion in text). The fold test reflects the principal component isolated for each sample ($N = 51$) taken from a block of sites in the upper part of the section before bedding tilt correction (C) and (D) after bedding tilt correction. A significant increase in precision in (D) relative to (C) indicates a positive fold test, where the characteristic component of magnetization was acquired before folding. The critical angle between the $N$ and $R$ populations is 5.3°. One outlier, sample 195.1, is considered spurious.

LEGEND
Limestone
Medium-fine grained to fine grained, poorly indurated sandstone
Fossil horizon
Shale
Fine grained, well-indurated sandstone
Fault
Basement
Siltstone
First appearance of Transverse Range clasts-transition from Mt. Eden to San Timoteo formation
Opdyke et al. classification
R N
Class I
Class III
Class IV
FLUVIAL DEPOSITION
LACUSTRINE DEPOSITION
Jackrabbit Trail
Shale faulted out
Continued
STRATIGRAPHIC HEIGHT (m)
VGP lat.
site 144
site 154
site 170
site 130
Mt. Eden Section 1
Mt. Eden Section 2
Mt. Eden Section 3
Frick Quarry

oteo data set is then dec. 347.7°, inc. 46.1° ($N = 82$, $\alpha_{95} = 2.8°$, $k = 32.2$).

A reversals test was performed to test the NRM stability [18,19]. Using all Class I sites the $\alpha_{95}$ cones of confidence for the normal and reversed populations do not overlap, indicating that the reversals test fails for the overall unit data set. However, if the block of nine reversed sites exhibiting large, seemingly anomalous, CW declination anomalies is removed, then the $\alpha_{95}$ cones of confidence overlap, indicating that these populations are statistically antipodal and, therefore, the reversals test (*sensu* [18]) is positive. With regard to the quality of this test, the critical angle between the normal and reversed populations is 5.3°, indicating a class B reversals test result [19]. In addition, a fold test [20] using the Class I site means for 51 samples from the upper part of the Jackrabbit Trail section (Fig. 5) indicate a significant increase in the grouping of the tilt-corrected data ($R = 48.4$, $k = 23.1$) into normal and reversed polarity populations (F ratio = 5.37, df = 50; $p > 0.001$) relative to the widely dispersed, non-tilt corrected data ($R = 39.3$, $k = 4.3$). These stability tests therefore indicate that the large majority of the characteristic magnetizations analyzed from the San Timoteo Badlands are stable and acquired before folding.

It should be noted that the sediments sampled from the Mt. Eden and San Timoteo formations are exposed in the north limb of an anticline that plunges about 10° to the northwest. Therefore, apparent declination anomalies attributable to the plunge of the bedding rotation axis must be considered. Following Chan [21], although inclination is unaffected, declination anomalies can be determined for given values of rotation axis plunge and rotation angle using:

$$\sin\delta = \frac{2\sin\mu\,\sin\theta}{1 + \sin^2\mu + \cos^2\mu\cos\theta}$$

where: $\delta$ = the declination anomaly; $\mu$ = plunge of the rotation axis; and $\theta$ = the angle of bedding rotation based on our field work. The rotation axis plunge ($\mu$) for the Mt. Eden and San Timoteo sediments is about 10° and the bedding dip ($\theta$) averages about 15–20°. Being conservative and using the greater value of 20° for ($\theta$), then the declination anomaly attributable to the plunge of the bedding rotation axis is $\leq -3.5°$ (negative sign indicates CCW). Correcting for this apparent declination anomaly, the Mt. Eden–San Timoteo mean direction is then dec. 351.2°, inc. 46.1° ($N = 82$, $\alpha_{95} = 2.8°$, $k = 32.2$).

## 4. Magnetic polarity stratigraphy and correlation to the MPTS

The site mean declinations and inclinations obtained using Fisher statistics [16] were used to calculate virtual geomagnetic pole (VGP) positions. The individual site VGP latitudes were then plotted against stratigraphic height for each of the JRT and Mt. Eden sections, resulting in a magnetic polarity sequence for the lower and middle parts of the San Timoteo Badlands (Fig. 6). The JRT section, continuous both above and below the lacustrine/fluvial depositional boundary, contains 13 polarity zones (N1–N3 and N5–R8). Near this depositional boundary a section of shale (represented in Mt. Eden section 1 and containing polarity zones R3, N4 and R4) has apparently been faulted out of JRT because, in this latter section, the shale bed is $< 10$ m thick while in the correlative Mt. Eden section the shale is $> 60$ m thick (Fig. 6). The four Mt. Eden sections, including the Frick quarry, contain a total of 8 polarity zones (R1–N5). Correlations within the San Timoteo Badlands sequence and to the MPTS are presented in Fig. 7. No volcanic units have been located in the San Timoteo Badlands deposits included in this study. Wherever possible, lithostratigraphic correlations were traced in the field using marker beds. Otherwise, it was necessary to use combinations of various factors, including lithological similarity, relative site positions, and fossil and paleomagnetic data, to assess correlations (also see

Fig. 6. Lithology and magnetostratigraphy for Jackrabbit Trail and the four Mt. Eden sections (including Frick quarry) with VGP (virtual geomagnetic pole) latitude plotted against stratigraphic height. Dominant lithologies, fossil horizons, and position of paleomagnetic sites are indicated. Note the approximate position of the change from lacustrine to fluvial depositional.

below). The composite reversal stratigraphy (Fig. 7) contains 16 polarity zones (N1–R8).

### 4.1. Temporal duration and pattern recognition

Given the fact that the sampling density in the San Timoteo Badlands (19.4 sites per expected reversal) is much greater than that necessary to sample adequately the section for magnetostratigraphy (it should be $\geq 8$ per polarity zone, according to Johnson and McGee [22]), then the number of reversals present can be used to estimate the temporal duration represented in a given paleomagnetic transect. Using Johnson and McGee [22], the amount of time, $\Delta t$, represented by a sampled section can be calculated using the formula:

$$\Delta t\ (\text{myr}) = S\tau N$$

where $S = -\ln(1-2p)/2$; $\tau$ = mean time span of polarity intervals; $N$ = the number of paleomagnetic

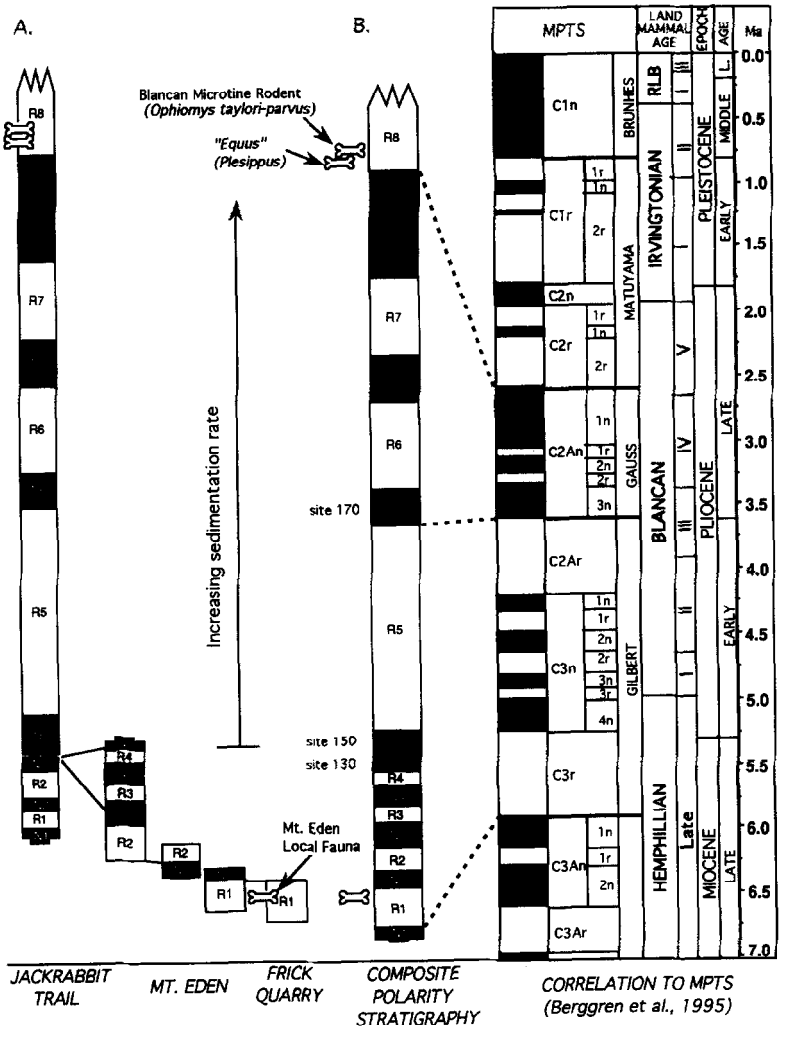



Fig. 7. (A) Magnetostratigraphy of Jackrabbit Trail and the four Mt. Eden sections with correlations between the sections and location of fossil horizons. (B) Composite magnetostratigraphy and correlation to the Magnetic Polarity Time Scale (MPTS) of Berggren et al. [10]. The San Timoteo sequence correlates to the MPTS from immediately below the Gilbert (chron C3An.1n) to the early Matuyama (chron C2r.2r) yielding an age range from about 6.0 to 2.5 Ma. North American land mammal ages are from Woodburne [24] and their boundaries are adjusted to the Berggren et al. MPTS [10].

sites; and $p = R/(N-1)$, where $R$ is the number of reversals encountered.

For the Mt. Eden and San Timoteo deposits sampled in this study, $N = 123$, $R = 15$ and thus $p = 15/122 = 0.123$ and $S = 0.141$. Using the MPTS of Berggren et al. [10], 17 polarity intervals occur between 2.0 and 6.0 Ma (the probable maximum age range of these sediments based upon fauna), yielding a mean interval length ($\tau$) of 0.23 myr. Substituting these values, the estimated amount of time required to deposit this part of the San Timoteo Badlands is 4.98 myr. However, since an overlap of sampling occurs at the base of the JRT section (zones N1–N3), the effective value of $N$ for the composite magnetostratigraphy is 108, rather than 123 sites. Recalculating using $N = 108$, the estimated amount of time

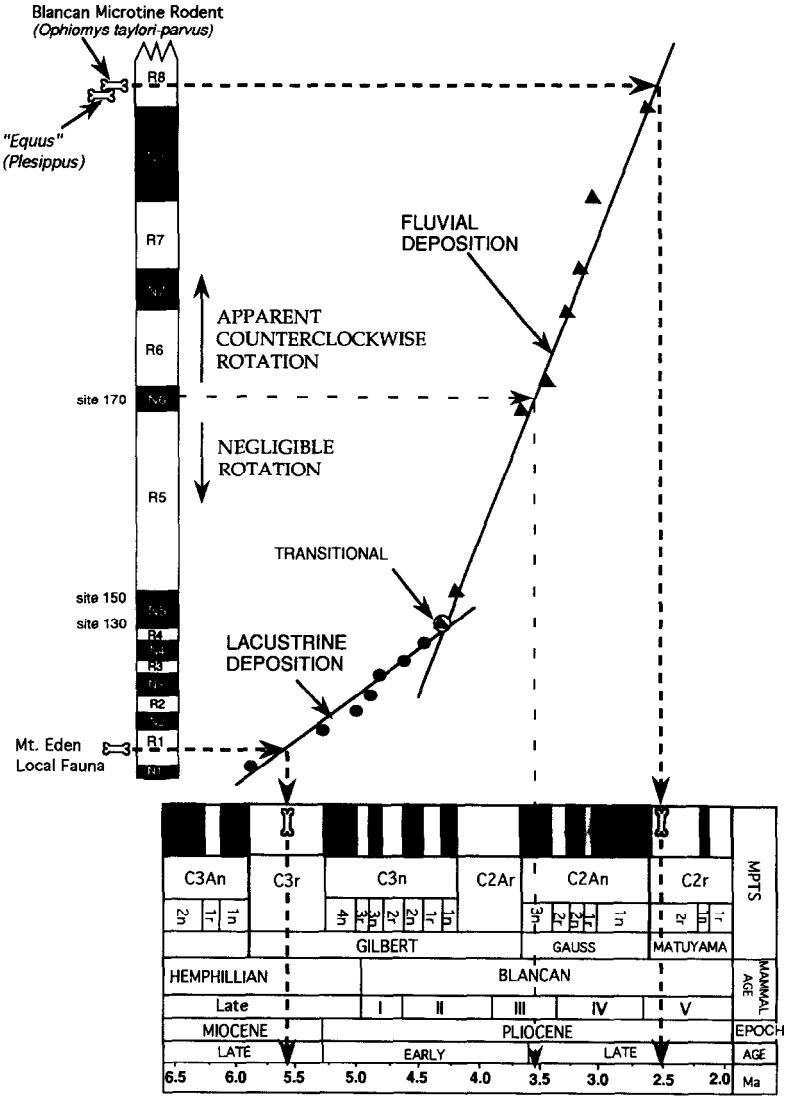



Fig. 8. Plot of composite magnetostratigraphy (left) against the MPTS [10]. Polarity transitions within the lacustrine regime are plotted as circles and those within the fluvial regime are plotted as triangles. A good linear fit is achieved for both lacustrine and fluvial depositional regimes, indicating a good correlation to the MPTS from about 6.0 to 2.5 Ma. Furthermore, the ages of the fossil horizons resulting from this correlation (about 5.6 Ma for the Mt. Eden fauna and 2.5 Ma for the occurrence of '*Equus*' (*Plesippus*) and the microtine rodent *Ophiomys taylori-parvus*) are consistent with predicted ages [1,9,11,23–25].

required for deposition is 4.0 myr. Thus, we choose this latter value for an independent estimate of the temporal duration of our paleomagnetic transect.

### 4.2. Biostratigraphy

Given the known age range for primitive representatives of the horse genus '*Equus*' (*Plesippus*) [11], the specimen collected near the top of the section sampled in magnetozone R8 is certainly younger than the Hemphillian–Blancan boundary at about 4.5 Ma [12], and could be as young as about 2.5–2.0 Ma. The Blancan IV to Blancan V microtine rodent (*Ophiomys taylori-parvus*, currently under study), also from magnetozone R8, should also be no younger than about 2.0 Ma [23]. The Mt. Eden L.F. occurs within the first reversed magnetozone (R1), near the base of the section. Based on the biostratigraphic presence of diagnostic late Hemphillian mammals and reversed magnetic polarity, May and Repenning [9] correlated this part of the Mt. Eden beds to the earliest reversed zone within the Gilbert Chron (chron 3r, *sensu* Berggren et al. [10]), with a revised age of about 5.5–6.0 Ma.

### 4.3. Correlation

Based upon temporal duration, faunal constraints, and overall polarity pattern, our composite section (Fig. 7) correlates from the top of chron C3An.1n to C2r.2r (about 6.0–2.3 Ma). Fig. 8 represents a plot of the composite magnetostratigraphy versus the MPTS [10]. A good linear correlation is achieved for both lacustrine and fluvial depositional regimes, indicating a good fit to the MPTS from about 6.0 to 2.3 Ma. Extrapolation from the composite section through the best-fit line to the MPTS yields an age of about 5.6 Ma for the Mt. Eden L.F. and 2.5 Ma for '*Equus*' (*Plesippus*) and the microtine rodent, *Ophiomys taylori-parvus*, consistent with previous ages from other lines of evidence [1,9,11,23–25].

## 5. Tectonic implications

### 5.1. Vertical-axis rotation

The expected North American paleomagnetic pole position for the late Miocene and Pliocene is essentially the same as the current pole position: dec. 360.0°, inc. 53.5° (for site lat. 34°N). The Mt. Eden–San Timoteo overall mean direction, after correcting for bedding tilt and for the declination anomaly (D) attributable to rotation axis plunge, is dec. 351.2° ± 4.0° (D ± ΔD, [26,27]), inc. 46.1°, suggesting tectonic rotation of the crustal block containing the San Timoteo Badlands. Relative to the stable North American continent, the San Timoteo Badlands in the study area indicate an overall vertical-axis rotation and error (R ± ΔR) of −8.8° ± 4.9° (negative sign indicates counterclockwise, CCW). A somewhat different result occurs if the formations are interpreted separately (Fig. 9).

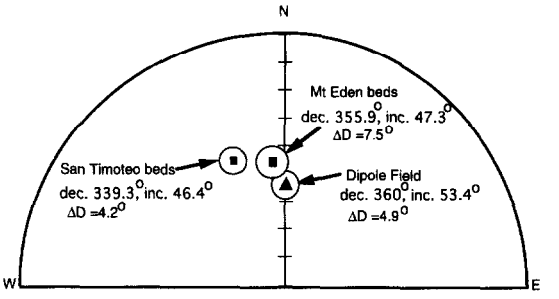



Fig. 9. Formation mean directions plotted on equal-area stereographic projection for the Mt. Eden and San Timoteo formations relative to the expected dipole field. Significant CCW rotation of −20.7° ± 5.2° is exhibited by the San Timoteo beds, while no significant rotation is observed for the Mt. Eden beds.

The lower part of the section containing the Mt. Eden deposits (below site 130; Fig. 6) has a mean (reversed sites inverted) of dec. 355.9° ± 7.5°, inc. 47.3°. The amount of rotation is negligible (i.e., R ± ΔR = −4.1° ± 9.1°). The middle part of the section, including sites 144–170, is difficult to evaluate because: (1) much of this interval is less fine-grained, yielding fewer Class I sites; and (2) the interval from site 159 to 167 exhibits anomalously large clockwise (CW) rotation. Overall, the reliable sites within this middle part of the section indicate negligible rotation. The upper part of the section (San Timoteo beds, site 170 and above) exhibits significant CCW rotation, with a mean of dec. 339.3° ± 4.2°, inc. 46.34°, yielding a rotation (R ± ΔR) of −20.7° ± 5.2°.

### 5.2. Implications

Results of the differential rotation analysis indicate that, while the basal Mt. Eden formation ex-

hibits little or no tectonic rotation, the upper San Timoteo formation, beginning at or near site 170 (about 3.5 Ma, Fig. 8), has undergone significant CCW rotation of $-20.7° \pm 5.2°$. Further, results of a fold test (Fig. 5) indicate that magnetization was acquired prior to folding. Hence, the CCW rotation of the upper JRT declination mean cannot be attributed to skewing of the overall declination mean of the reversed sites by normal overprints. These results suggest that a decoupling of the upper and lower parts of the section must have occurred between about 3.5 and 4.0 Ma, allowing independent vertical-axis rotation of the upper San Timoteo sediments. Although independent, small-block rotation is plausible, and has been noted in other paleomagnetic studies in tectonically active regions [27,28], to date, no field structural evidence of any decoupling has been found in the San Timoteo Badlands.

To determine if the presence of significant outliers might be the cause of the CCW rotation of the upper JRT declination mean, a plot of site declinations against stratigraphic height was constructed (Fig. 10). In Fig. 10A, the negative slope of the best-fit line suggests differential CCW rotation within the section. As can be seen by comparing Fig. 10A and B, removal of all significant site declination outliers has little effect on the slope of the best-fit line. This argues that the differential counterclockwise rotation cannot be attributed to outliers.

The behavior of each sample during stepwise thermal demagnetization was analyzed. For a few reversed sites above site 176 in the top part of the JRT section, at temperatures above about 350°C, the direction of magnetization changes slightly (between 5° and 10°). This behavior suggests that these site declinations may have been skewed slightly by hard, secondary overprints carried by hematite. However, there are so few of these sites that their removal has no significant effect on the declination mean for the upper San Timoteo deposits.

### 5.3. Flattening

The overall inclination (reversed polarity sites inverted) for the Class 1 sites from San Timoteo Badlands is 350.9°, inc. 46.2° ($N = 91$, $\alpha_{95} = 3.1°$, $k = 27.4$). The mean of normal sites is dec. 346.8°, inc. 41.8° ($N = 40$, $\alpha_{95} = 4.1°$, $k = 35.6$) and the mean of all reversed sites is dec. 354.8°, inc. 50.4° ($N = 51$, $\alpha_{95} = 4.2°$, $k = 28.0$). Using the dipole formula ($I_x = \tan^{-1}$ [2 tan $\lambda$], where $I_x$ = expected inclination and $\lambda$ = present-day geographic latitude [27]) for this latitude (34°), the expected inclination should be 53.5°. The overall mean inclination of 46.2° for the San Timoteo Badlands is shallower than would be predicted from the dipole formula [27]. There are several possible reasons for this inclination shallowing (or flattening; $F \pm \Delta F = 7.3 \pm 1.6°$

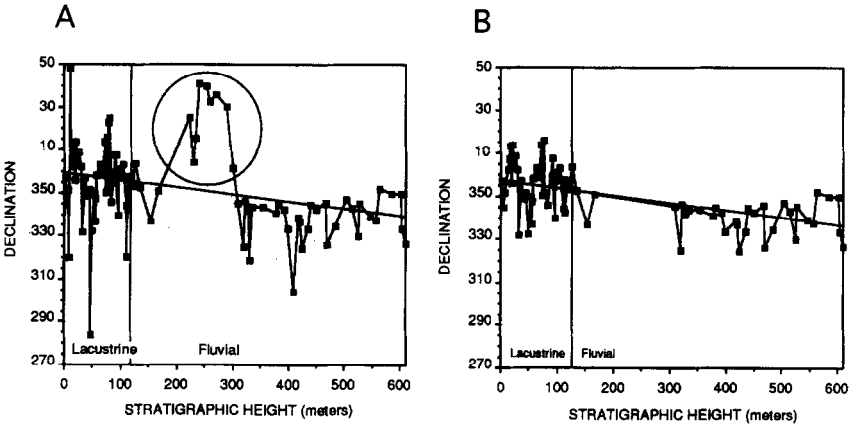



Fig. 10. Plot of declination (reversed rotated to northern hemisphere) against stratigraphic height. (A) All Class I sites. A block of nine reversed sites exhibiting large CW declination anomalies probably attributable to local deformation is circled. The negative slope of the best-fit line suggests possible differential CCW rotation within the section. (B) Class I sites with significant outliers removed. The slope of the best-fit line does not change significantly, indicating that the apparent differential CCW rotation is not due to outliers. The mean declinations for the lacustrine and fluvial parts of the section are, respectively, about 358°and 341°.

[26,27]), also called inclination error. The most plausible seems to be post-depositional compaction, which is a phenomenon frequently observed in paleomagnetic studies of sedimentary rocks [27].

## 6. Summary and conclusions

Based upon faunal constraints, temporal duration, and overall polarity pattern, the composite magnetostratigraphy for the San Timoteo Badlands, which includes the Mt. Eden and San Timoteo formations, is correlated to the MPTS from Chrons C3An.1n to C2r.2r (ca. 6.2–2.3 Ma; Figs. 7 and 8). Extrapolation from the composite section through the best-fit line to the MPTS yields an age of about 5.6 Ma for the Mt. Eden L.F. and 2.5–2.2 Ma for the occurrences of '*Equus*' (*Plesippus*) and the microtine rodent *Ophiomys taylori-parvus* in the upper part of the JRT section.

The overall mean for the San Timoteo Badlands, corrected for bedding tilt and for apparent declination anomalies attributable to rotation axis plunge, is dec. $351.2° \pm 4.0°$. This suggests slight, or negligible, CCW rotation of the fault block containing the San Timoteo Badlands. At a smaller scale, if the formations are interpreted separately as subunits, it is possible that differential vertical-axis small-block rotation occurred. The Mt. Eden beds, with a formation mean of dec. $355.9° \pm 7.5°$, inc. 47.3°, exhibit negligible CCW rotation of $-4.1° \pm 9.1°$. Conversely, the upper San Timoteo beds (site 170 and above) have an overall mean of dec. $339.3° \pm 4.2°$, inc. 46.34° and therefore exhibits significant CCW rotation of $-20.7° \pm 5.2°$. No structural evidence of decoupled, vertical-axis small-block rotation has yet been found in the field.

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